

A GHG inventory for a firm that does not exist

The estimand

Total greenhouse-gas emissions, in tCO₂e, for reporting year 2024, of an illustrative office-based financial-services firm with offices in Frankfurt, Paris and Milan, by scope, with a 95% Monte Carlo interval on every number.

Why illustrative

The firm is invented and says so on every page. What is not invented is any fact about Germany, France or Italy: emission factors (DESNZ GHG conversion factors 2024, full set v1.1), grid intensity, office energy intensity and commuting patterns are all fetched from public endpoints by `src/emissions_inventory.py`. The firm's own headcount, commute distance and travel budget are assumptions, labelled as such, in `results/emissions_inventory.json:assumptions` -- never fetched, because no public source describes a firm that does not exist.

The answer

Location-based: 1,070 tCO₂e [950, 1,194] across 450 employees.

Market-based: 1,491 tCO₂e [1,354, 1,630] -- 39% higher than location-based.

The method a firm picks for Scope 2 changes the headline number more than any other choice in this inventory. Page 3 shows why.

Method: boundary, sources, vintages

Boundary

GHG Protocol Corporate Standard, operational control. Reporting year 2024.

Offices: Frankfurt (200 employees), Paris (150 employees), Milan (100 employees).

Activity data

Office electricity and gas: Eurostat final energy consumption in commercial and public services (nrg_bal_c, FC_OTH_CP_E), divided by Eurostat services-sector employment (nama_10_a10_e) to give a national per-employee intensity, scaled by the firm's assumed headcount per city.

Emission factors, fetched, vintages recorded

Scope 1 and Scope 3 combustion/travel: DESNZ GHG conversion factors 2024, full set v1.1.

Scope 2 location-based: Eurostat env_air_gge (CRF sector 1.A.1.a, public electricity and heat production) / nrg_bal_c (gross electricity production) -- replicates EEA's own stated method; EEA's CSV endpoint returned HTTP 410 when this build ran.

Scope 2 market-based: AIB European Residual Mixes 2024, CO2 sheet.

all three source families carry reporting-year-2024 data; no vintage mismatch found, which is itself a checked outcome, not an assumption.

Result: by scope, with an interval on each

Reconciliation

Location-based total 1,070 tCO₂e vs market-based

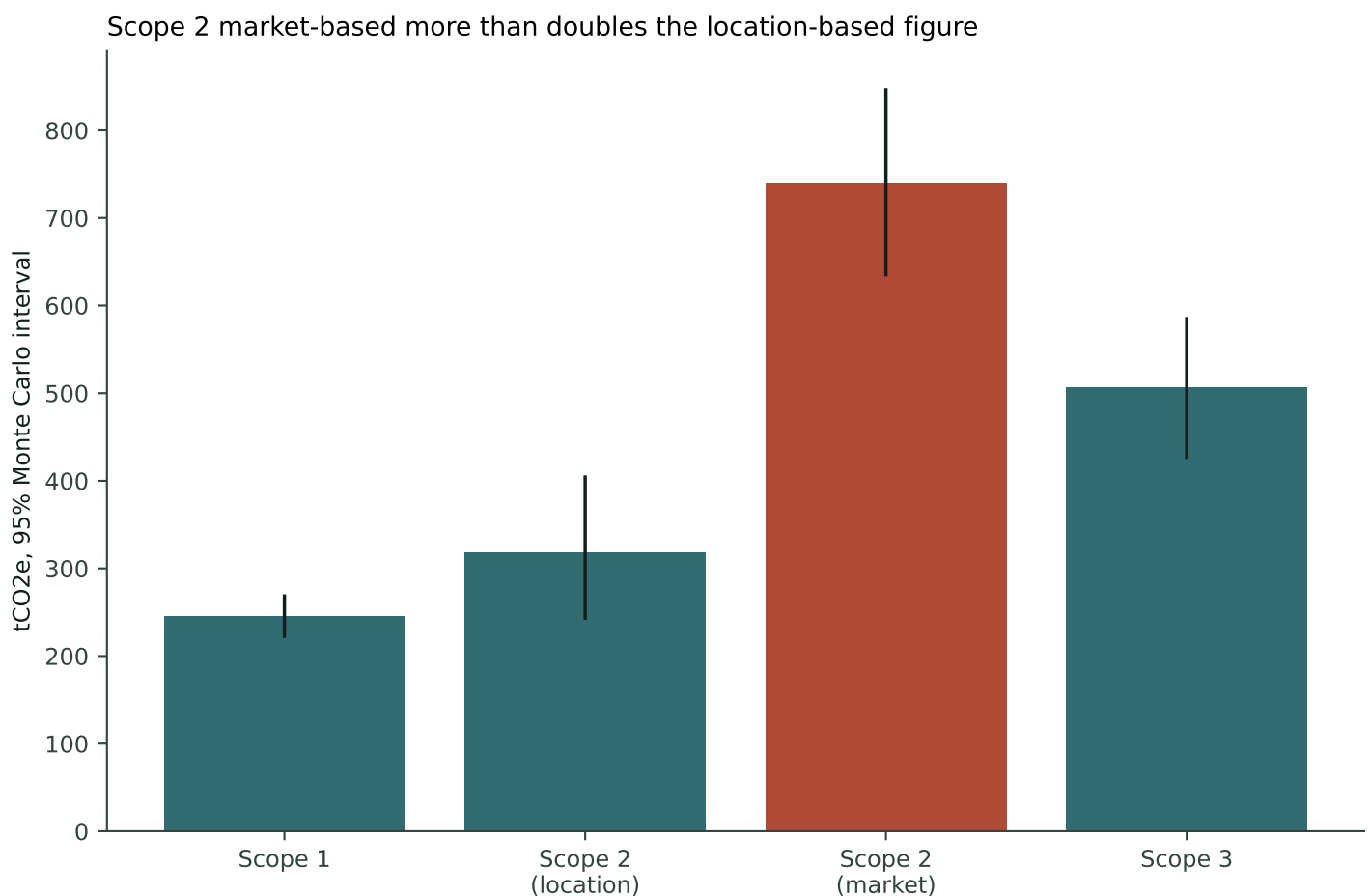
1,491 tCO₂e, a gap of 421 tCO₂e

(39%). The gap is Germany: its AIB residual mix

is priced near-fully fossil because German generators sell most of their renewable attributes as Guarantees of Origin, leaving the 'residual' pool almost entirely grey. A firm buying no certificates reports against that residual mix.

What drives the interval

Top 3 variance contributors: Frankfurt Purchased electricity, market-based (sd=52 tCO₂e); Frankfurt Purchased electricity, location-based (sd=41 tCO₂e); Frankfurt Commuting, car (sd=31 tCO₂e).



Validation against a source this module did not fit

The check

AIB publishes its own 'production mix CO2' figure per country alongside the residual mix this module uses for Scope 2 market-based -- computed by Grexel on Ecoinvent data, independent of the Eurostat CRF1A1A/GEP ratio this module fits for the location-based figure. The two methods should agree in direction even though they are not the same calculation.

What it shows

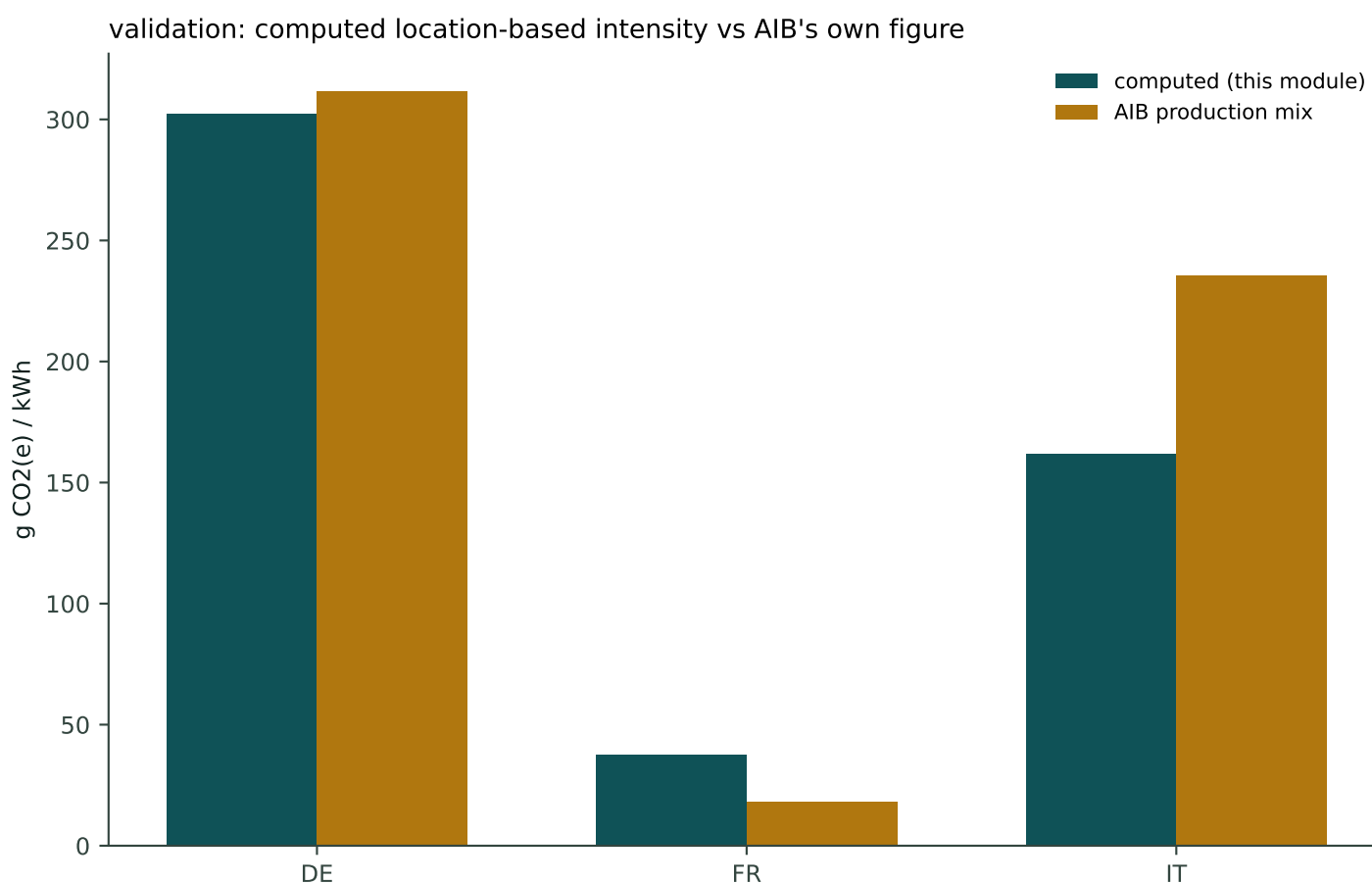
Germany: 3% apart -- close agreement.

France: 107% apart --

France's computed figure (CRF1A1A/GEP) is roughly double AIB's Ecoinvent-based figure; both are near zero in absolute terms (French generation is >90% nuclear and hydro), so a small absolute gap reads as a large percentage one.

Italy: 31% apart --

CRF1A1A excludes autoproducer (industrial CHP) generation, which is material in Italy; this is the gap this module's own method note flags as a limitation.



Sensitivity: the choices nobody can justify alone

Scope 2 method: the biggest mover

Location-based 1,070 tCO₂e vs market-based 1,491 tCO₂e -- a 39% swing from one methodological choice the GHG Protocol requires firms to report both sides of, precisely because neither is obviously right.

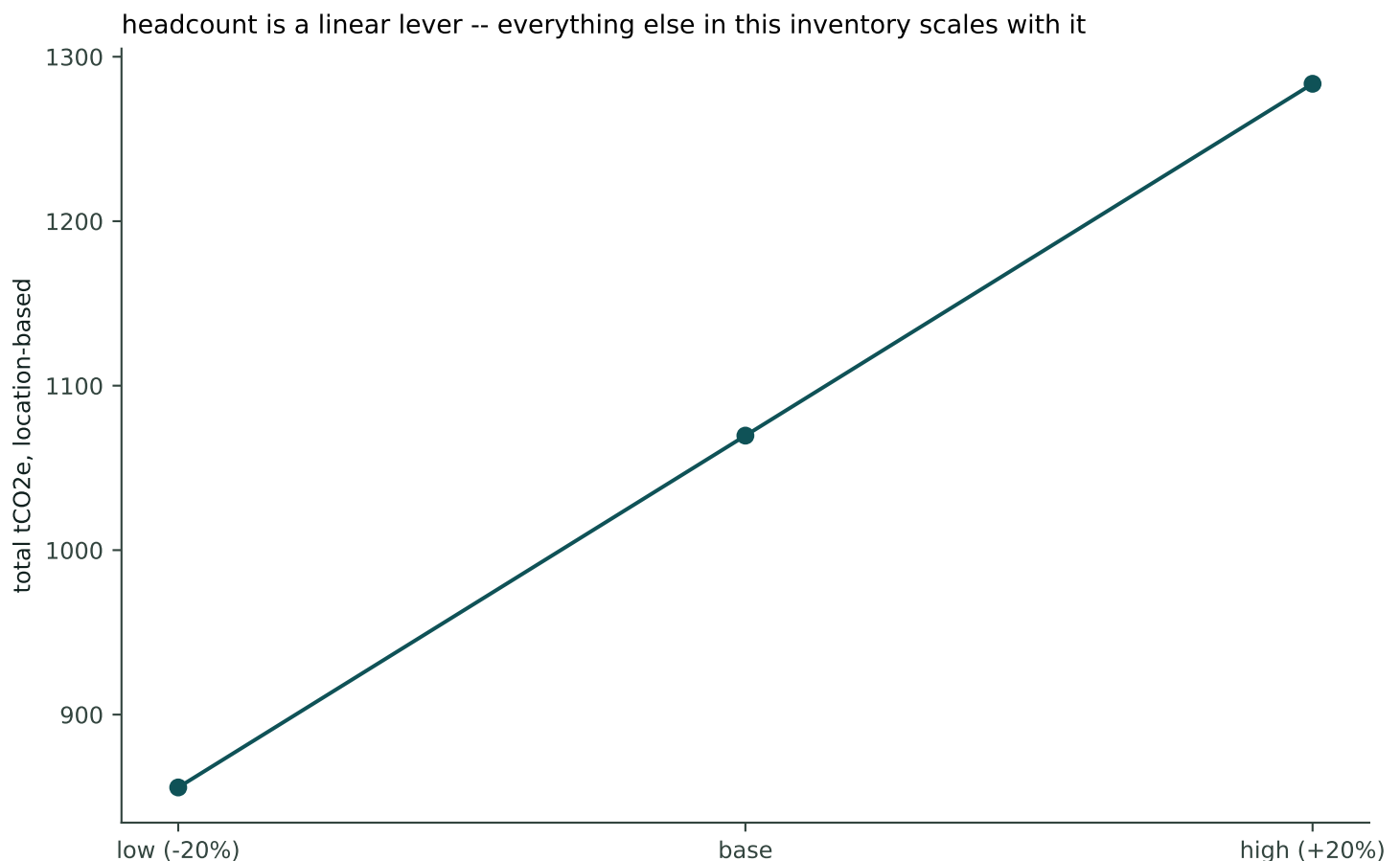
Calorific value convention: a small but real trap

Correct (Net CV, matching Eurostat's activity basis): 245 tCO₂e.

Gross CV factor misapplied to the same net-basis activity: 221 tCO₂e (-10%). Small next to the Scope 2 method choice, but the kind of error that passes a sanity check.

Scope 3 boundary

Categories 6+7 (business travel, commuting) are 41% of the location-based total -- material, and built on this module's least constrained assumptions (commute distance, travel budget). Dropping them would understate the inventory by exactly that share.



Limits, and what I would build next

The assumption that breaks the conclusion

Every activity-data line in this inventory is a national sector average scaled by an assumed headcount, not a metered bill. If this firm's actual electricity or gas intensity per employee differs from the national commercial-services average by more than the stated uncertainty range, every downstream number moves with it -- the Monte Carlo interval reflects the assumption's stated range, not a verified bound on how wrong the assumption could be.

Scope 3 completeness: 3 of 15 categories built

Category 1 (purchased goods and services) is the largest likely gap: no public spend-based emission-factor set is reachable with this module's library stack (stdlib, numpy, pandas, matplotlib, openpyxl only); likely the largest true gap for an office-based services firm (IT services, professional services, office supplies).

Full completeness table with every exclusion's reason:
results/emissions_inventory.json:completeness.

No backtest, and why not one for this module

STANDARD.md requires backtesting anything forward-looking. Nothing in this module is a projection -- it is a single historical reporting year (2024) built from data already realised, not a forecast of a future one. There is no held-out year to test a forecast against, because this module makes no forecast.

Next

Module 05 adds Category 15 (financed emissions) for a real, disclosed portfolio -- the category this module excludes and names as the reason.

Order-of-magnitude check: 1.25 tCO₂e/employee (Scope 1+2 location-based), within the sanity bound [0.3, 6.0]: True.